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Predicting employee absenteeism: A framework and its validation

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ABSTRACT

The issue of employee absenteeism presents a continual obstacle to the effectiveness of organizations, especially within sectors that rely heavily on labor. This research introduces an innovative probabilistic framework that employs Bayesian Belief Networks (BBNs) to forecast the risk of absenteeism as well as the absenteeism rate, tackling the intricate and uncertain relationships among various contributing factors. By employing thorough factor analysis and drawing on expert insights, we systematically identify and organize essential variables spanning personal, organizational, familial, health-related, and external domains. The model integrates the insights of experts and addresses the challenge of limited data by employing a functional interpolation technique to develop conditional probability tables. Two models based on Bayesian Belief Networks have been developed: one assesses the probability of an employee being absent on a specific day, while the other predicts absenteeism rates on a monthly basis. The Monte Carlo simulation serves as a valuable tool for addressing the uncertainty inherent in essential input variables. The validation process, which includes scenario, sensitivity, and diagnostic analyses, highlights the strength and relevance of the models. This study presents a practical and scalable resource for decision-makers to evaluate and address absenteeism, particularly in contexts where data may be limited.

1. Introduction

The issue of employee absenteeism remains a pressing concern for organizations around the world, as it has a direct impact on productivity, operational efficiency, and the overall stability of the workforce. Frequent or prolonged absences disrupt workflows, diminish morale, and create financial challenges [1]. Consequently, tackling absenteeism is essential for sustaining performance and ensuring effective human resource planning [2]. Absenteeism is influenced by a complex interplay of personal, organizational, environmental, and psychosocial factors. However, current models frequently overlook the element of uncertainty, do not adequately represent the interconnections among these factors, and struggle to operate effectively when data is scarce [3].

Earlier research emphasises elements like individual health, familial responsibilities, job contentment, and workload; however, it frequently approaches absenteeism in an overly simplistic manner. Many methods tend to be deterministic and depend significantly on extensive, well-organised datasets, which restricts their use in situations where data may be incomplete, inconsistent, or qualitative in nature. Moreover, they tend to neglect nonlinear interactions, which diminishes the practical utility of traditional statistical models.

To address these limitations, this research employs Bayesian Belief Networks (BBNs), which are particularly useful in situations characterized by uncertainty and limited data. Two models have been created: one that estimates the risk of daily absenteeism and another that forecasts monthly rates. Both approaches integrate personal, organizational, family, health, and external influences, employing factor analysis to refine the data while maintaining its explanatory strength. Expert elicitation plays a crucial role in the development of conditional probability tables, with the assistance of functional interpolation to alleviate the demands placed on experts. Monte Carlo Simulation (MCS) enhances the reliability of predictions by evaluating the impact of uncertainty.

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This research is framed around three key questions.

- RQ1: What are the primary factors that affect absenteeism, and in what ways can they be organised into systematic categories?
- RQ2: In what ways can probabilistic modeling be utilized to predict absenteeism risk and rates when faced with uncertain or incomplete data?
- RQ3: In what ways does uncertainty influence predictions of absenteeism, and what methods can be employed to quantify this through simulation?

This research distinguishes itself from previous studies by incorporating a diverse range of factors rather than concentrating on limited categories. Insights from experts play a crucial role in forming connections in areas where empirical data is limited, allowing for the anticipation of absenteeism risks at both the organizational level and for specific tasks.

The primary goals are to.

- Identify the most significant factors contributing to absenteeism,
- Develop models based on Bayesian Belief Networks to predict the risk and rate of absenteeism,
- Establish causal relationships among these factors.

This research offers valuable insights through the development of two predictive Bayesian Belief Network models, which assist supervisors in evaluating the likelihood of daily attendance and the average absence rates for individuals or specific sections. Factor analysis facilitates the tailoring and strength of models, whereas Monte Carlo Simulation enhances realism through the integration of uncertainty. An accessible expert elicitation method enhances the practicality of subjective Bayesian Belief Network (BBN) applications, particularly in contexts where data is limited.

The subsequent sections of this paper are organised in the following manner: In Section 2, we will explore the existing literature; Section 3 will outline the essential methods employed; Section 4 will detail the development of the model with numerical illustrations; Section 5 will illustrate the implementation process with sensitivity analysis and highlight the key findings; Section 6 will engage in a discussion of the results; and the concluding section 7 will reflect on the limitations encountered and propose directions for future research.

2. Literature review

Different studies have held employee absenteeism responsible for significantly hurting productivity, quality, efficiency, and profitability of organizations. The bulk of the research on absenteeism has focused on identifying the factors responsible for its occurrence. For the convenience of the readers, all the factors of absenteeism identified in this literature review are listed in Table 1.

Research done on absenteeism prediction can be broadly divided into two distinct sectors.

1 Testing the significance of different factors as predictor of absenteeism
2 Predictive Modeling of absenteeism

The majority of the studies reviewed worked on the first research category. However, the second category of research has seen a significant spark of interest in recent times.

Merekoulias & Alexopoulos [35] sought to assess the characteristics of the work ability index (WAI), the Bradford formula (index) or factor (BF), and their combination in predicting future sickness absence. Mukasa [15] tried to investigate causes for absenteeism among health employees in rural Uganda. Data were gathered employing a demographic questionnaire and focus groups [13]. introduced four tests for selecting important causes of absenteeism- Kruskal-Wallis test, Mann-Whitney test, Spearman Correlation Analysis and Univariate Analysis. And recently Cochran-Armitage Test was used for checking linearity [7]. In recent times, researchers have attempted to integrate artificial intelligence into the modeling of absenteeism. One of the first such attempts was [36]'s application of a neuro-fuzzy network to predict the absenteeism of courier company personnel. Another model was proposed by Ref. [37], which used a recurrent neural network to predict turnover intention of new employees at restaurants. Machine learning algorithms have also been implemented to predict employee absenteeism in recent studies [38], [39], [40], [41].

BBNs have been popularly applied in research due to their ability to capture complicated inferences [42]. In another research [43], a novel method, a Dynamic Bayesian Network (DBN) pipe crash forecast model

Table 1
Causes of absenteeism.

Variables	Reference
Anxiety	[4]
Psychological Empowerment	[5]
Stress	[6]
Work Ability	[7]
Quality of Health	[8]
Diseases	[7], [4]
Accident	[9]
Alcohol Consumption	[7]
Smoking	[7]
Age	[4]
Gender	[10], [11], [12]
Level of Education	[4]
Marital Status	[13]
Family Size and Members	[14]
Astrological Sign	[13]
Domestic Violence	[15]
Tenure	[13]
Family Issues	[6]
Personality	[16]
Debt	[17]
Salary	[15]
Alternate Earning	[15]
Payment Delay	[15]
Household Income	[18]
Commitment	[6]
Perceived Absence Norm	[19]
Perception of Justice	[20], [21]
Intention to Leave Organization	[22]
Job Satisfaction	[23]
Job Security	[6]
Organizational Trust	[24]
Work Unit Size	[25]
Promotion Opportunity	[26]
Leadership/Supervision	[15], [27]
Ownership Structure	[28]
Task Variety	[26]
Task Repetitiveness	[26]
Autonomy	[23]
Job Type	[4]
Work Type	[13]
Working Hour	[6]
Job Complexity	[26]
Job Monotony	[6]
Role Clarity	[26]
Task Clarity	[26]
Work Engagement	[29]
Physical Workload	[30]
Mental Workload	[30]
Peer Interaction	[4]
Relation with Boss	[31]
Noise	[32]
Transportation	[15]
Infrastructure	[15]
Working Condition	[6]
Time Pressure	[23]
Country of Origin	[33]
Unemployment Rate	[34]

that acknowledges various static, dynamic, and operational pipe failure factors for pipe failure forecast was introduced. Drury [44] assumed that it is arguable that Bayesian Networks (BN)s are specially fitted to agricultural study because of their ability to reason with insufficient data and incorporate new information. Afrin & Yodo [45] used BN for the probabilistic estimating of traffic congestion. Adedipe [46] presented a systematic analysis of existing research on the application of BN models in the wind energy sector. Ibrahim [47] used BN for the system-level dependability evaluation for high power light-emitting diode light study. Steijn [48] did research on integrating human factors into quantitative risk analysis utilizing BBNs towards developing a Quantitative Risk Analysis (QRA).

Of late, researchers in the field of human resource management have shown interest in using BBN models. Chakraborti [49] used a BBN model to predict expected salary class of employees based on their performance. On the other hand [50], introduced a model to evaluate employee performance.

Several limitations were identified in absenteeism research during the literature review. Firstly, in the reported models, there is a lack of representation of all categories of factors like personal, organizational commitment, family-related, health related, organization related, extrinsic, and their joint effects. Since the relationship between the variables is non-linear and complex, therefore, expert opinion is also required to express those relationships. Moreover, most of the reported models were designed to be deterministic, making them ineffective to deal with uncertainty related to important parameters such as income opportunity, sick leave availability, transportation availability, weather and accident in real-word scenario. Since parameters can be both qualitative and quantitative, data integration considering uncertainty of related parameters is also a major issue that needs to be addressed. Secondly, extraction of Conditional Probability Table (CPT) was found to be the biggest hindrance to developing a subjective BBN model. The existing elicitation methods to build CPT from limited initial information were found to be challenging for the experts. Therefore, simplification of the elicitation process is required. Finally, absenteeism has been linked with numerous variables. BBN has the challenge of discretizing continuous variables, receiving and forming expert experience and knowledge, etc. [51]. Developing the model would, therefore, require both dimensionality reduction and rendition maximization. However, guideline for such modeling has been discussed by very few studies. This research tried to address all these questions and suggested effective solutions for them.

3. Methodology and model development

The objective of this research is to develop models for predicting the absenteeism rate of employees in the workplace. This research proposes two models to predict absenteeism.

1 Model to Predict Absenteeism Risk at a Particular Day
2 Model to Predict Absenteeism Rate in a Fixed period

The first model is applicable for individual employees, while the second is applicable for both individual and group of employees. For developing these two models, the steps in Fig. 1 are carried out sequentially. The last step with MCS is applicable for the first model only.

3.1. *Identification of important factors causing absenteeism using factor analysis*

Initially, 57 factors were chosen as causes of absenteeism from the existing literature given in Table 1. Next, their applicability in the workplaces of Bangladesh was verified by surveying the local working population. 169 people employed in diverse roles at different organizations and industries were asked 57 questions (one for each factor) regarding the significance of these factors. The survey was done both face to face and online. The questionnaires of the survey are provided in Appendix A as ***supplementary materials*** and the responses were taken on a linear scale-from complete disagreement to full agreement. The survey was done on both paper and using google form.

After the completion of the survey, the suitability of the received data was checked using Kaiser-Meyer-Olkin (KMO) test, Bartlet's sphericity test, and determinant. IBM SPSS 22 software was used for all the statistical analysis. The result of that analysis can be seen in Table 3 under the column “initial stage”. Though the results met the minimum requirement, the KMO value was not satisfactory. To improve the value, factors were removed arbitrarily and the consequent impact on the KMO value was observed. After conducting multiple trial and errors, the KMO value was raised to 0.725 at the discretion of 16 factors. The result is shown in the “After Data Screening” column of Table 2.

With the data adequacy ensured, factor analysis was run on the reduced dataset of 41 factors. The parameters for the analysis are given in Appendix B as ***supplementary materials***.

Based on eigenvalue, Scree plot (Fig. 2) detected 14 variables incorporating the 41 factors explaining 66.92% of the total variance

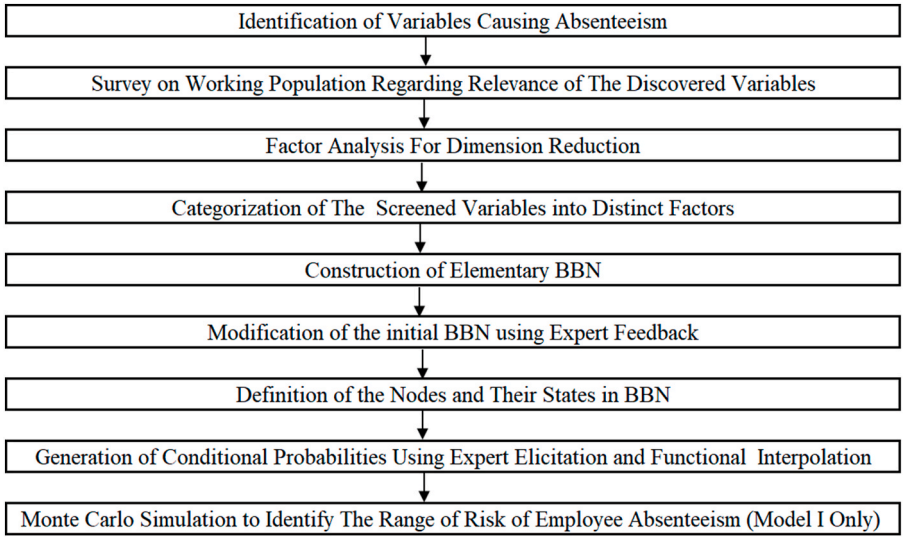


Fig. 1. Steps involved in developing model to predict absenteeism.

Table 2
Indicators of data suitability.

Criteria	Initial Stage	After Data Screening
No of factors	57	41
KMO value	0.639	0.725
Significance in Bartlett's test of sphericity	0	0
Determinant	3.49×10^{-11}	5.32×10^{-7}

(provided in Appendix B as *supplementary materials*). Table C3 in Appendix C shows Factor loading of the factors over each variable which indicate that how much the factor serves as a pure measure of the variable. Considering the base for the inclusion of a factor into the absenteeism model as loading of 0.5 for any factor, the final tally of significant factors shrank to 32.

3.2. Theoretical foundation for variable selection

The phenomenon of employee absenteeism, while often viewed through the lens of statistics or management, is fundamentally rooted in the behavioral aspects that reflect an individual's cognitive, motivational, and contextual decision-making processes [3]. Theories derived from behavioral and organizational psychology, such as Absence Theory, the Theory of Planned Behavior (TPB), and Social Cognitive Theory (SCT), are recognized for providing substantial conceptual foundations that elucidate the reasons behind individuals' decisions to attend or withdraw from work. The intentional, normative, and situational dimensions of absence behavior are collectively explained by these perspectives, which are captured through probabilistic interactions among personal, organizational, health, family, and environmental variables within the Bayesian Belief Network (BBN) framework.

3.2.1. Absence theory and withdrawal behavior

Absence theory is understood as a perspective that characterizes absenteeism as a manifestation of withdrawal behavior, whereby employees are seen to temporarily disengage from unfavorable elements of their job or organization [53]. Grounded in the research of job attitudes, it is proposed that the costs and benefits of attending work versus being absent are evaluated by employees. It has been observed that when dissatisfaction, stress, or exhaustion surpasses the perceived rewards, an increase in the likelihood of withdrawal is noted [54]. It is acknowledged that factors such as job satisfaction, workload, organizational justice, and supervisory relationships are central to this theoretical lens. In the present study, variables such as job satisfaction, payment delay, role clarity, promotion opportunity, and supervision quality are utilized to represent these constructs.

The inclusion of these elements within the organizational-commitment and organizational-impact domains is aligned with the assertion of Absence Theory that absenteeism functions as a coping or adjustment mechanism in response to work strain.

3.2.2. Theory of Planned Behavior (TPB) [55]

It has been suggested by the Theory of Planned Behavior that any deliberate behavior, such as workplace attendance, is influenced by behavioral intention. This intention is, in turn, shaped by three key determinants: the attitude toward the behavior, subjective norms, and perceived behavioral control.

- The attitude toward attendance is reflected in the evaluation made by an employee regarding the decision to attend work or to remain at home. The attitudinal component is represented by variables such as job satisfaction, stress, mental workload, and physical workload. The likelihood of absenteeism is increased by a negative work attitude, which is often amplified by poor working conditions or high strain.
- Subjective norms are understood as the perceived social pressure that influences an individual's decision to attend or be absent, which arises from the expectations held by supervisors, peers, or family members. Within the framework, it is observed that factors such as the quality of supervision, organizational justice, penalties for absence, and family support serve to operationalize these normative pressures.
- Perceived behavioral control is understood to include the perceived ease or difficulty associated with attending work, which is influenced by various situational enablers or barriers. These may include factors such as the availability of transportation, prevailing weather

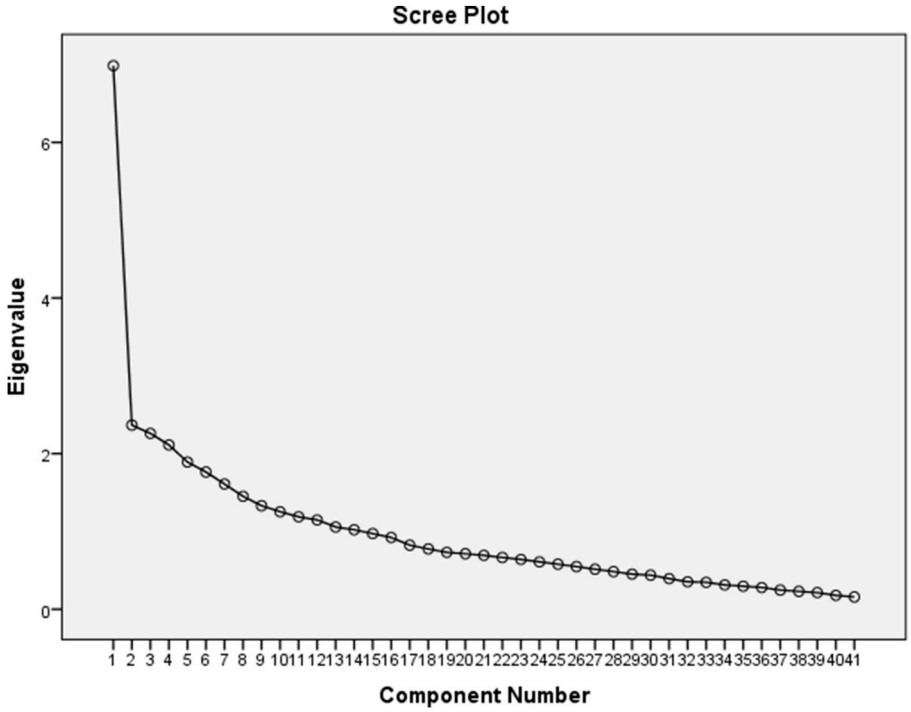


Fig. 2. Screen plot of factor analysis.

conditions, individual health status, the distance to the workplace, and family obligations. The extrinsic and health-related domains of the BBN are incorporated with these elements.

The mapping of these psychological constructs onto measurable organizational and personal variables allows for the capturing of how intentions to attend (or not attend) evolve under conditions of uncertainty. The relationships are allowed to be represented as probabilistic dependencies by the BBN, reflecting the continuous interplay that exists between individual cognition and external conditions.

3.2.3. Social Cognitive Theory (SCT) [56]

The SCT is seen as a complement to the TPB, with an emphasis placed on reciprocal determinism, which refers to the dynamic interaction that occurs among personal factors, behavioral patterns, and environmental influences. From this perspective, it can be understood that absenteeism is not solely the result of static traits (e.g., age, gender), but rather emerges from continuous feedback loops involving individual capability, motivation, and situational reinforcement. It has been observed that variables such as work ability, anxiety, family support, organizational climate, and infrastructure exemplify these interdependencies.

In the BBN structure, relationships are operationalized through feedback-like linkages among personal, health, family, and organizational nodes, which allows for the quantification of how environmental conditions may reinforce or mitigate tendencies toward absenteeism over time.

The integration of Absence Theory, the Theory of Planned Behavior, and Social Cognitive Theory is ensured to provide a foundation for variable selection that is grounded in behavioral rationale and summarized in Table 3.

Consequently, the 32 factors identified through factor analysis are systematically associated with well-established constructs that elucidate the cognitive, affective, and contextual drivers of absenteeism. Theoretical foundations are embedded within a probabilistic framework, enabling the BBN to model not only static determinants but also the behavioral dynamics of attendance decisions in the context of uncertainty. The integration of the methodological rigor associated with industrial engineering and the explanatory strength found in behavioral science is observed, resulting in a holistic and theoretically informed model of employee absenteeism.

3.3. Categorization of the factors

Expressing absenteeism risk as a direct function of 30 factors would make the model extremely complicated. Even considering the simplest network with each node having two discrete states, the total number of conditional probabilities required would be $2^{32} = 4294967296$, which is nearly impossible. To avoid this issue, divorcing [57] was employed. From literature, eleven factors were identified to have relation with organizational commitment. To separate the rest, five mediate factors were introduced. In another word, the existing factors were separated into 6 separate categories. Table 4 lists these categories and the causes falling under them.

In the first model any of the six factors can trigger absenteeism on a particular day. However, absenteeism rate is predicted over a certain period rather than a single day. Therefore, the effect of those factors would not be the same as in model I. That is why two new major determinants of absenteeism rate were introduced, which are Disturbance Frequency and Motivation. Disturbance frequency denotes the frequency of events leading to absenteeism while motivation denotes the general interest of a person in skipping his/her office. The higher their value, the greater the absence rate would become.

3.4. Bayesian Belief Network (BBN)

BBN is a Bayes' theorem-based method that deals with the probabilistic assessments of problems involving a large number of variables. What today is referred to as Bayes' theorem, Bayes' equation, or the Law of Inverse Probability, is achieved by solving $P_r = \{A \mid BX\}$:

$$P_r\{A|BX\} = \frac{P_r\{A|X\}\, P_r\{B|AX\}}{P_r\{B|X\}} \tag{1}$$

In the above, the logical propositions are A, B, and X, with AB describing the joint statement; A|BX defining the conditional statement A, given B and X are true.

It is a directed acyclic graph (DAG) where variables of interest are presented as nodes and the conditional dependencies among the variables are represented by the edges or links. The quantitative relationships among the connected nodes in a BBN are represented as Conditional Probability Distributions (CPDs) for each variable in the network. They are determined either from data or by expert judgment or both [58]. In cases of data scarcity, researchers often have to obtain CPDs using expert elicitation.

Generally, three types of nodes can be observed in a BBN, that is, (1) Root Node (without parent nodes), (2) leaf nodes (without child nodes), and (3) intermediate nodes (nodes with parent and child node). An illustrative example of BBN is demonstrated in Fig. 3.

In a BBN analysis, let us assume there are n variables D1, D2, D3, … … Dn where Z = {D1, D2, D3, … … Dn} is the set of conditionally independent variables. The dependency of node D2 on node D1 can be shown by an arc that is directed outwardly from D1 to D2. So, it can be said that the D2 is D1's child and the other way around. A CPT is developed to quantify the dependencies. Unconditional probabilities (UP) are defined for nodes that do not have parents. The full joint probability distribution of BBN using the chain rule can be expressed as:

$$P(D_1, D_2, \ldots .. D_n) = P\left(D_1 \left| D_2 D_3, \ldots D_n\right.\right) P(D_2|D_3, D_4, \ldots D_n) \ldots P(D_{n-1}|D_n) P(D_n) \tag{2}$$

Equation (2) can be represented as:

Table 3
Mapping of behavioral constructs to representative variables and their corresponding BBN domains.

Behavioral Construct	Representative Variables in This Study	BBN Domain
Attitude toward work	Job satisfaction, stress, workload	Organizational commitment/health impact
Subjective norms	Supervision, justice perception, family support	Organizational impact/ Family impact
Perceived control	Transportation, health, distance, infrastructure	Extrinsic/Health-related
Withdrawal tendency	Job monotony, payment delay, promotion limitation	Organizational commitment
Reciprocal environment	Weather, infrastructure, organizational size	Extrinsic/Organizational impact

Table 4
Categorization of factors.

Category of the Factors	Factors
Personal	Age, Gender, Education Level, Personality, Sick Leave, Alternate Income Opportunity
Organizational Commitment	Job Complexity, Working Hour, Task Variety, Work Involvement, Mental Workload, Physical Workload, Role Clarity, Payment Delay, Promotion Opportunity, Job Satisfaction, Perceived Justice
Family Related	Family Size, Children, Family Support, Marital Status
Health-Related	Work Ability, Stress, Anxiety, Accident
Organization Related	Supervision, Organization Size, Environment, Penalty for Absence
Extrinsic	Infrastructure

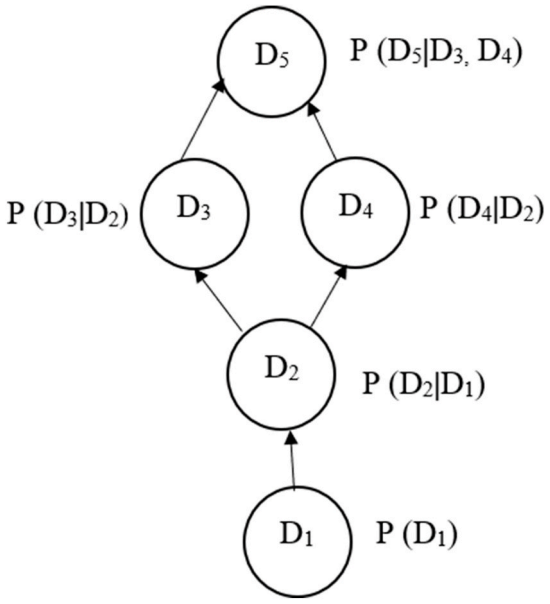


Fig. 3. An illustrative example of BBN. D_1 is root nodes, D_2, D_3 and D_4 are intermediate nodes and node D_5 is a leaf node [52].

$$P(D_1, D_2, \ldots..D_n) = \prod_{i=1}^{n} P(D_i|D_{i+1}, \ldots.D_n) \quad (3)$$

The joint probability distribution of BBN expressed in Eq. (3) can be evaluated based on the information of the parent nodes. For instance, if D_4 is the child node of both D_2 and D_3, then the part of joint probability distribution can be put back with P ($D_4|D_2$, D_3). Therefore, the full joint probability distribution of BBN will be:

$$P(D_1, D_2, \ldots..D_n) = \prod_{i=1}^{n} P(D_i|parents\ (D_i)) \quad (4)$$

3.5. Construction of initial BBN structure

With the selection and categorization of the factors completed, the initial BBN for the two models can be constructed. The construction consists of three steps, which are Defining Nodes, Defining the states of the Nodes, Defining Links.

At the initial stage, Model I consist of 37 nodes whereas Model II due to additional categorization consists of 39 nodes. They have been described in Table 5.

Table 6 lists all the links between nodes used in the two models.

3.6. Modification of the initial model

Two modifications were made to the model based on expert feedback.

- Merging the node child and marital status
- Addition of parent nodes to extrinsic impact node

Both the nodes' "Children" and "Marital Status" are parents of "Family Impact" with two and three states respectively. Among the six possible combinations between them, *unmarried with children* is not feasible in the context of our country whereas *unmarried without children* is redundant. Therefore, to reduce computational expense, the two nodes were merged into a single node "Marital Status" with four states: Single, Married, Married with children and Single with Children.

On the other hand, only one factor was listed under extrinsic impact. Therefore, upon consultation with the experts, *weather* and *distance from work* nodes were introduced as parent nodes to extrinsic factors. Moreover, to clarify measurement of infrastructure, two-parent nodes were added: road quality and transportation availability.

3.7. Calculation of conditional probability

The last step in absenteeism modeling consists of the formulation of CPT, which denotes the conditional relationship between different factors. Due to insufficient available data, the models had to rely on the subjective opinion of experts. Again, while obtaining these opinions, two

Table 5
Weight and Influence for parent nodes of different aspects.

Factors	Average Weight	Influence	Factors	Average Weight	Influence
Health Impact			Infrastructure		
Accident	5.43	0.169	Transportation Availability	1.55	0.480
Work Ability	8.01	0.249	Road Quality	1.67	0.520
Stress	8.72	0.271	**Motivation**		
Physical Workload	10.04	0.312	Personal Impact	4.22	0.218
Organizational Impact			Organizational Commitment	5.91	0.306
Supervision	10.41	0.32	Organizational Impact	9.19	0.476
Organization Size	3.22	0.1	**Disturbance Frequency**		
Workplace Environment	10.13	0.3	Extrinsic Impact	4.16	0.215
Penalty for Absence	9.44	0.28	Family Impact	8.19	0.424
Family Impact			Health Impact	6.97	0.361
Gender	5.82	0.181	**Extrinsic Impact**		
Family Size	5.91	0.184	Weather	8.71	0.451
Marital Status	7.59	0.236	Distance	5.49	0.284
Family Support	12.88	0.4	Infrastructure	5.12	0.265
Organizational Commitment			**Absenteeism Rate**		
Job Satisfaction	7.02	0.218	Disturbance Frequency	6.44	0.670
Justice Perception	7.66	0.238	Motivation	3.22	0.330
Promotion Opportunity	6.64	0.206	**Absenteeism**		
Payment Delay	10.88	0.338	Personal Impact	14.18	0.210
Job Satisfaction			Organizational Commitment	13.77	0.204
Task Variety	7.92	0.164	Organizational Impact	15.54	0.230
Work Involvement	8.90	0.184	Extrinsic Impact	5.17	0.076
Mental Workload	11.98	0.248	Family Impact	10.66	0.158
Physical Workload	6.76	0.140	Health Impact	8.30	0.123
Role Clarity	12.74	0.264			

Table 6
List of links with reference.

Child Node	Parent Node	Source
Personal Impact	Age	Contribution to This Research
	Gender	
	Education Level	
	Personality	
	Alternate Income Opportunity	
	Sick Leave Availability	
Mental Workload	Job Complexity	Contribution to This Research
	Working Hour	[59]
Physical Workload	Working Hour	[60]
Job Satisfaction	Task Variety	[61]
	Work Involvement	[62]
	Mental Workload	[61]
	Physical Workload	[61]
	Role Clarity	Contribution to This Research
Organizational Commitment	Justice Perception	[63]
	Job Satisfaction	Contribution to This Research
	Promotion Opportunity	Contribution to This Research
	Payment Delay	[64]
Family Impact	Family Size	Contribution to This Research
	Marital Status	
	Children	
	Gender	
	Family Support	
Health Impact	Stress	
	Physical Workload	
	Work Ability	
	Accident	
Stress	Anxiety	[65]
	Mental Workload	[66]
Organization Impact	Penalty for Absence	Contribution to This Research
	Supervision	
	Organization Size	
	Workplace Disturbance	
Extrinsic Impact Disturbance Frequency	Infrastructure	
	Health Impact	
	Family Impact	
	Extrinsic Impact	
Motivation	Organizational Commitment	Contribution to This Research
	Organizational Impact	
	Personal Impact	
Absenteeism	Personal Impact	
	Organizational Impact	
	Health Impact	
	Personal Impact	
	Extrinsic Impact	
	Organizational Commitment	
Absenteeism Rate	Disturbance Frequency	
	Motivation	

major challenges were encountered. The first one was the quantification of the CPT. Subject experts may easily provide qualitative opinions about the possibility of different outcomes, but they are not comfortable with providing a specific value. Secondly, the total number of conditional probabilities required for the two models was 599 and 642 respectively. Extracting such a large amount of information from experts would be time-consuming. To resolve these problems, this research employs two solutions.

- •Elicitation of a limited number of conditional probabilities (anchor) from expert opinions using rank-based method
- •Calculation of the remaining conditional probabilities using Functional Interpolation method.

The whole process for deriving the conditional probabilities can be divided into the following steps.

- •Selection of the experts and evaluation of their credibility.
- •Data Collection from Experts.
- •Assessment of the influence of state change in each parent node on the child node.
- •Calculation of the anchor CPD from these influences.
- •Approximation of the anchor CPDs as normal function and determination of parameters.
- •Calculation of the parameters of the intermediate CPDs using linear interpolation.
- •Calculation of the remaining conditional probabilities by converting the parameters into discrete probability distributions.

All of the steps are explained in detail in **Appendix C** in the ***supplementary materials***. The complete CPT table for infrastructure is mentioned in Table 7.

With the CPT formulation complete, the two models are finalized and ready for application. The final structure of Model I and Model II are shown in Figs. 4 and 5.

4. Numerical illustration

For testing the models, data were collected for 14 employees in a particular section of a renowned ready-made garment (RMG) factory of Bangladesh. Individual interviews, organizational policy and supervisor's opinion (Personality, Work Ability, Promotion Opportunity etc.) were used to obtain both individual and organizational data. Data like age, gender, education level, marital status, family support, distance from work, and sick leave availability of the employees, environment, task variety and etc. are available in Appendix D of the ***supplementary materials***.

For testing the models, we contacted a renowned multinational RMG manufacturer operating in Bangladesh. The company has been functioning for over 35 years with 11 manufacturing facilities across Asia and Africa. We collected data from its facility in Dhaka. In particular, we focused on the sampling section, which usually involves production of prototypes prior to large-scale manufacturing. The section consisted of 14 employees. Most of the information were collected through individual interviews. Others were obtained from organizational policy and supervisor's opinion (e.g., personality, work ability, promotion opportunity). This information is summarized in Table Individual interviews, organizational policy and supervisor's opinion (Personality, Work Ability, Promotion Opportunity etc.) were used to obtain both individual and organizational data. Appendix G of the ***supplementary materials*** denotes these data.

Table 7
Complete CPT of infrastructure.

State of Parent Nodes		CPD of Infrastructure			Parameters of Corresponding Normal Function	
Transportation Availability	Road Quality	Poor	Average	Good	Mean	Standard Deviation
High	Poor	0.52	0.35	0.13	1.61	0.71
High	Mediocre	0.01	0.70	0.29	2.31	0.36
High	Excellent	0.00	0.00	1.00	3.00	0.00
Medium	Poor	0.71	0.29	0.00	1.31	0.36
Medium	Mediocre	0.19	0.78	0.03	1.83	0.37
Medium	Excellent	0.01	0.65	0.34	2.35	0.38
Low	Poor	1.00	0.00	0.00	1.00	0.00
Low	Mediocre	0.66	0.34	0.00	1.35	0.38
Low	Excellent	0.48	0.35	0.17	1.69	0.75

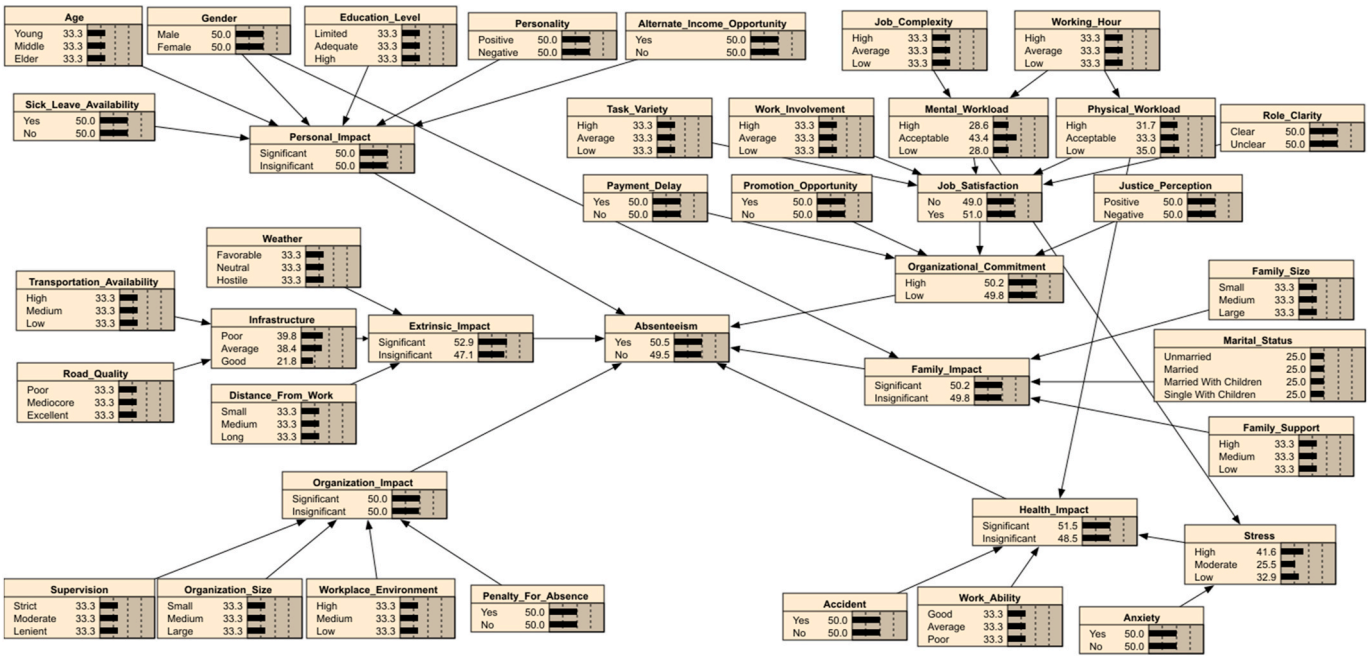


Fig. 4. Finalized version of Model I.

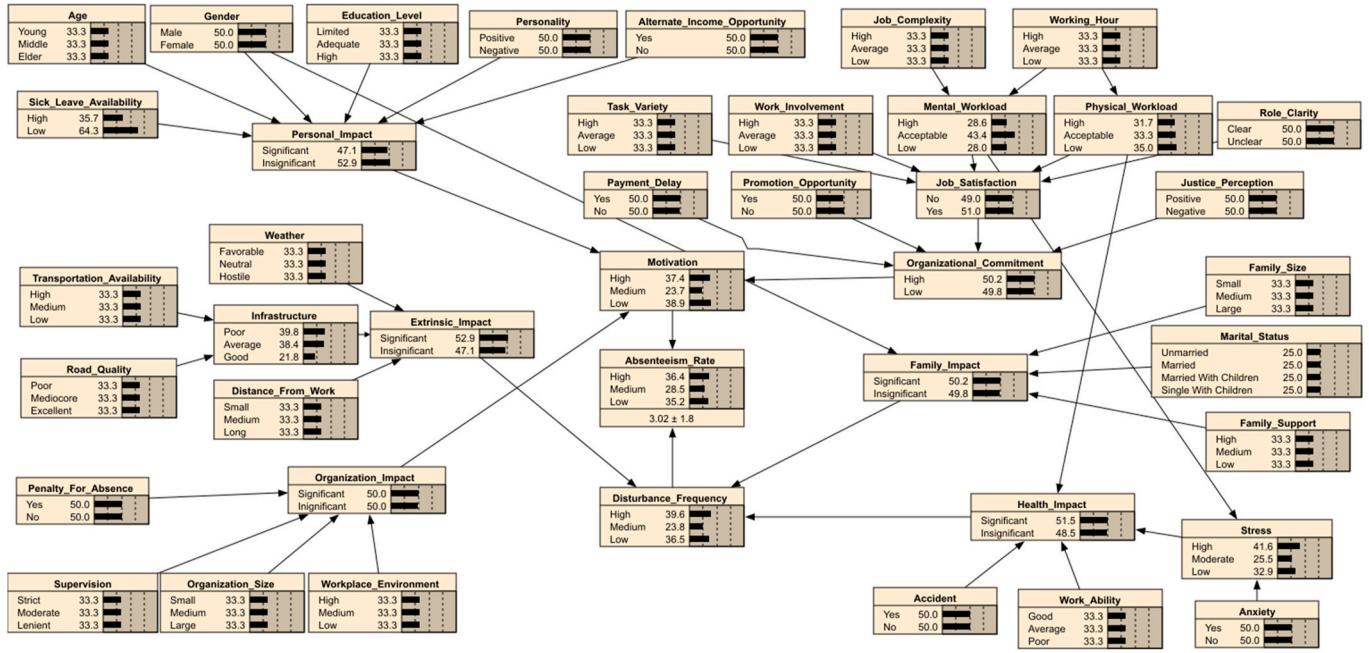


Fig. 5. Finalized version of Model II.

4.1. Organization-wise data

Summarizing the data found in Appendix D of the ***supplementary materials*** would help determine the mean absenteeism rate of the organization. Table 8 provides this summary regarding the states of different attributes.

4.2. Defining the high, medium and low rate of absenteeism

Attendance data of the employees over previous one year (January 2019-December 2019) were collected to identify the rate of absence per month. Fig. 6 summarizes the data.

From the data, the first and third quartile were identified to be 2 and 4 respectively. As per previous definition, the range of the three absenteeism rates was defined in Table 9.

4.3. Scenario generation for Monte Carlo Simulation (MCS)

To observe the effect of uncertainty on individual absenteeism risk, five uncertain factors were randomly selected for simulation, which are Alternate Income Opportunity, Sick Leave Availability, Transportation Availability, Weather and Accident. Using the probability distribution of these factors, 100 scenarios were generated using MCS.

Table 8
Organization-wise attribute data.

Attribute	States	Prior Probability (%)	Attribute	States	Prior Probability (%)
Age	Young	78.6	Gender	Male	64.3
	Middle-aged	21.4		Female	35.7
Personality	Positive	50.0	Alternate Income Opportunity	Yes	35.7
	Negative	50.0		No	64.3
Education Level	Limited	57.1	Job Complexity	High	7.1
	Adequate	28.6		Average	21.4
	High	14.3		Low	71.4
Sick Leave Availability	High	35.7	Role Clarity	Clear	85.7
	Low	64.3		Unclear	14.3
Working Hour	High	35.7	Task Variety	High	7.2
	Average	50.0		Average	21.4
	Low	14.3		Low	71.4
Work Involvement	High	85.7	Family Size	Small	71.2
	Average	0.0		Medium	28.8
	Low	14.3		Large	0.0
Family Support	High	28.6	Work Ability	Good	35.7
	Medium	28.6		Average	50.0
	Low	42.8		Poor	14.3
Workplace Environment	High	14.6	Transportation Availability	High	14.3
	Medium	70.8		Medium	78.6
	Low	14.6		Low	7.1
Justice Perception	Positive	92.9	Anxiety	Yes	78.6
	Negative	7.1		No	21.4
Road Quality	Poor	35.7	Distance from Work	Small	35.7
	Mediocre	64.3		Medium	42.9
	Excellent	0.0		Large	21.4
Marital Status	Single	28.6	Accident	Yes	7.1
	Married	21.4		No	92.9
	Married with Children	42.9			
	Single with Children	7.1			

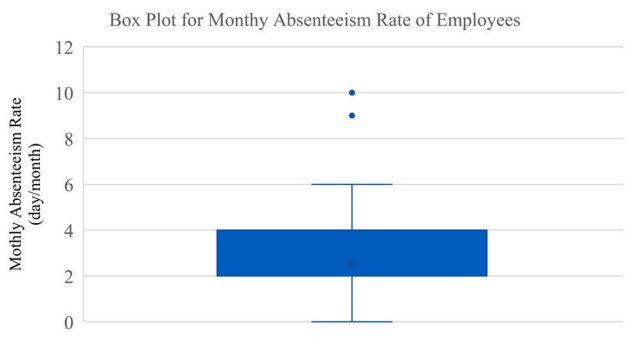


Fig. 6. Box plot of monthly absenteeism rate.

Table 9
Definition of different states of absenteeism rate.

State	Absenteeism Rate per Month
High	>4
Medium	2-4
Low	<2

4.4. Testing of models

The following tests were done using the data accumulated in the previous sections.

- Individual and common information were provided to Model I as prior probabilities to calculate the absenteeism risk for each employee.
  •Absenteeism risk under uncertainty was calculated for each employee on Model I using the scenario generated by MCS.
  •Individual and common information were provided to Model II as prior probabilities to compute the absenteeism rate for individual employees.
  •Organization-wide information was provided to Model II as prior probabilities to compute the mean absenteeism rate for the whole section.

For implementation of these tasks, Netica software was used.

5. Analytical results

The second column of Table 10 shows the absenteeism risk of employees on any particular day derived from Model I. The third to fifth column shows the risk under stochastic conditions in three forms-highest, lowest and mean risk. The last two columns correspond to the mean absenteeism rate of employees computed through Model II. Table 11 shows the probability distribution of the organization-wide employee absenteeism rate.

Therefore, the mean absenteeism rate per month would be:

$$\mu = (5 \times 0.239) + (3 \times 0.265) + (1 \times 0.496)$$

$$= 2.49 \text{ days}$$

And the standard deviation was calculated as 1.7 days.

5.1. Model validation

Four techniques were used for testing the validity of the models. The results of the tests are shown in the following sections.

5.1.1. Extreme condition test

This test evaluated how the absenteeism risk and absenteeism rate model fared in case of the most favorable and least favorable situation. Table 12 highlights the results of most favorable condition (Extreme 1) and least favorable condition (Extreme 2). It indicates that the probability of absenteeism rate is low (99.4%) and the mean absenteeism rate

Table 10
Absenteeism risk and mean absenteeism rate of individual employee.

Employee No	Risk (%)	Under Stochastic Condition			Mean Absenteeism Rate (Days per Month)	Standard Deviation
		Mean Risk (%)	Highest Risk (%)	Lowest Risk (%)		
1	34.7	39.9	51.8	33.1	2.50	1.7
2	43.1	46.8	64.8	39.9	2.78	1.8
3	26.8	28.6	41.2	21.3	2.45	1.7
4	44.7	45.9	59.5	38.9	3.02	1.8
5	43.3	46.9	59.3	40.0	2.69	1.8
6	26.5	29.7	42.2	23.4	2.11	1.6
7	33.5	38.1	51.6	30.5	2.32	1.7
8	30.1	32.7	45.0	25.7	2.36	1.7
9	21.1	23.7	35.3	18.0	2.01	1.6
10	38.9	41.8	55.5	30.3	2.55	1.7
11	53.9	54.9	66.8	47.6	3.28	1.8
12	37.5	38.7	50.5	32.7	2.58	1.8
13	17.7	21.1	33.7	14.5	1.84	1.5
14	24.6	29.3	42.1	22.3	2.09	1.6

Table 11
Probability distribution of employee absenteeism rate in the organization.

Absenteeism Rate	High	Medium	Low
Probability (%)	23.9	26.5	49.6

is only 1.02 days for Extreme 1 condition while for Extreme 2 condition, the absenteeism rate is high (99.4%) with a mean absenteeism rate of 4.98 days. Therefore, the developed model can capture the two extreme conditions effectively.

5.1.2. Scenario analysis

Three factors were chosen for implementing it randomly, which are alternate income opportunity, payment delay and anxiety. Different combinations of the parameters are considered to generate various scenarios, and their probability distribution is tested to perform model validation. However, it could not be shown here due to space limitations. Table 13 shows five scenarios where the prior probability of alternate income opportunity, anxiety and payment delay was varied from 90% to 10%. The results of the scenario analysis for both models are given in Table 14.

5.1.3. Sensitivity analysis

Sensitivity analysis was applied to both models in two ways. The first one identified the sensitivity of the output to individual factors, while the second one identified the sensitivity of the output to the factors introduced in the model. For Model I, entropy-based sensitivity analysis was used due to the discrete state of absenteeism. Using entropy-based sensitivity analysis has the benefit that they do not require a particular design of experiments. They can be applied to simulations resulting from other analyses [67]. For Model II, however, both variance-based and entropy-based sensitivity analysis were applicable. The results are shown in Figs. 7–10.

For Employee absenteeism, Weather revealed the highest contribution (36.2%) for the entropy reduction followed by Physical_workload (28.7%), Stress (26.6%), Working_hour (22.8%), Mental_Workload (19.1%), and Distance_from_Work (16.6%). To a lesser degree, Infrastructure, Penalty_for_Absence, Work_Ability, Family_Support, Accident, Supervision, Gender, Job_Satisfaction, and Payment_Delay exhibit sensitivities of 12.9%, 12.8%, 12.6%, 11%, 11%, 10.8%, 10.8%, 10.6%, and 8.8%, respectively. Finally, Anxiety, Justice_Perception, Workplace_Environment, Road_Quality, Marital_Status, Transportation_Availability, Promotion_Opportunity, Alternative_Income_Opportunity, Family_Size, Job_Complexity, Sick Leave_Availability, Organization_Size, Personality, Age, Role_Clarity, Education_Level, Work_Involvement, Task_Variety showed lower sensitivity in the range of 4.4% to 0.04%.

Therefore, in child nodes, based on Fig. 9, Disturbance_Frequency has the most contribution (22.4%) for the entropy reduction. After that, Motivation (10.1%) is in the second place. Health_Impact (2.94%), Extrinsic_Impact (2.29%), Organization_Impact (1.46%), Family_Impact (1.1%), Organization_Commitment (0.83%), and Personal_impact (0.52%) revealed lower sensitivities. Hence, for example in here, the overall absenteeism of employees is highly dependent on the type of Weather. The result of this examination also revealed that the sensitivity of the child node highly depends on the variability of the parent nodes.

5.2. Key findings

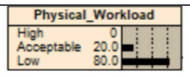
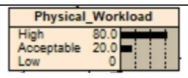
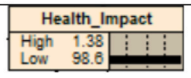
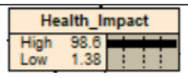
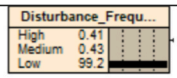
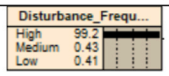
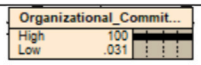
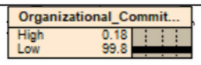
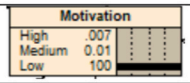
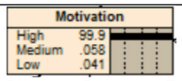
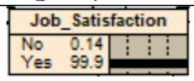
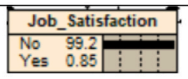
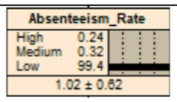
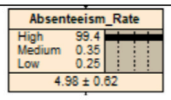
Several findings have been noted from the results of model tests and validation tests. Firstly, the employee absenteeism risk was found to be greater under stochastic conditions than under deterministic conditions. However, in both cases, it was less than 50% for all except one employee. Employee 13 is predicted to have the least risk while employee 11 is predicted to be most vulnerable to absenteeism.

Secondly, Model II predicted the mean absenteeism rate of individual employees as well as that of the whole section. In the case of individual tests, only one employee has been predicted to have a mean absenteeism rate under 2 days per month. 3 are expected to skip office for more than 3 days, while the rest are supposed to have an absenteeism rate between 2 and 3 days. The employees with the highest and lowest absenteeism rate were employees 11 and 13 respectively, which showed consistency with the results of model I. On the other hand, the result of the section-wise test has two interpretations. The first one is the mean absenteeism rate in the section which was determined 2.49 days per month. From another perspective, it has also quantified the proportion of the working population having high, medium and low absenteeism. In this case, the numbers are 23.9%, 26.5%, and 49.6%, respectively. This allows organizations to get a summary of absenteeism without conducting individual analysis.

Thirdly, sensitivity analysis appraised the effect of both the variables and the factors on the models' outcome and distinguished the most influential ones. In the case of Model I, weather has been identified as the most prominent factor while among them, health impact was the most influential one. The same results were found for Model II.

Finally, the validity tests have demonstrated satisfactory results for both models. In the Extreme-condition test, both models performed according to expectation. The analysis from 5 scenarios did not show any inconsistency in the model outputs either. And diagnostic analysis confirmed that the probability of the factors' state conducive to absenteeism decrease as the absenteeism risk shifts from 100% to 0%. The same observation was noted for model II. Therefore, the proposed models can be stated as valid.

Table 12
Extreme-condition test of the suggested BBN-based result framework.

Parent node	Extreme 1	Extreme 2		Extreme	Extreme
Education_Level	High	Limited	Gender	Male	Female
Sick_Leave_ Availability	Low	High	Personality	Positive	Negative
Personal_Impact	Insignificant	Significant	Age	Elder	Young
Job_Complexity	Low	High	Working_Hour	Low	High
Work_Involvement	High	Low	Task_Variety	High	Low
Mental_Workload	Low	High	Role_Clarity	Clear	Unclear
Alternate_Income_ Opportunity	No	Yes	Transportation_ Availability	High	Low
Promotion_Opportunity	Yes	No	Payment_Delay	No	Yes
Distance_From_ Work	Small	Long	Road_Quality	Excellent	Poor
Justice_perception	Positive	Negative	Family_Size	Small	Large
Organization_Size	Small	Large	Family_Impact	Low	High
Workplace_Environment	High	Low	Family_Support	High	Low
Penalty_for_Absence	Yes	No	Marital_Status	Unmarried	Single
Extrinsic_Impact	Insignificant	Significant	Work_Ability	Good	Poor
Physical_ Workload			Stress	Low	High
Health_ Impact			Accident	No	Yes
Disturbance_ Frequency			Anxiety	No	Yes
Organizational_ Commitment			Supervision	Strict	Lenient
Motivation			Weather	Favorable	Hostile
Job_Satisfaction			Infrastructure	Good	Poor
Absenteeism Rate					

6. Discussion on the findings

This study aimed to tackle the ongoing issue of predicting employee absenteeism by presenting a framework based on Bayesian Belief Networks (BBN) that incorporates diverse inputs and openly represents uncertainty. The results confirm that absenteeism is a multifaceted phenomenon shaped by personal, organizational, familial, health-related, and external factors. The results indicate that probabilistic techniques more effectively capture interdependencies and yield actionable insights that deterministic models frequently overlook.

Previous studies have identified a wide array of factors that contribute to absenteeism. Health-related factors, including stress, anxiety, and work ability [4], [16], as well as organizational aspects such as job satisfaction, supervisory relationships, and perceptions of justice [31], consistently emerge as significant predictors of absenteeism. The responsibilities associated with family life, such as caregiving duties and the number of family members, have been emphasized in recent studies [15]. The current findings are in strong agreement with existing literature, as the BBN models reliably pinpointed these categories as significant factors affecting absenteeism risk and rates. Nonetheless, a significant progression in the research has been achieved through the demonstration of the nonlinear interactions among these various factors. It was observed that weather conditions, physical workload, and stress were significant factors in sensitivity analyses; however, it was noted that their impacts were intensified when evaluated in conjunction with family support and organizational justice. The capacity of the model to quantify interactions across various domains is distinguished from previous methods, which were often characterized by the analysis of predictors in isolation [13].

Another important distinction is found in the manner in which uncertainty is addressed. Although traditional statistical and regression-based models are recognized for their valuable contributions, it has

Table 13
Scenario analysis of the suggested BBN-based result framework.

Scenario	Parent node			
	Alternate Income Opportunity	Anxiety	Payment Delay	Absenteeism Rate
Scenario 1	Alternate_Income_Opportunity: Yes 90.0, No 10.0	Anxiety: Yes 90.0, No 10.0	Payment_Delay: Yes 90.0, No 10.0	Absenteeism_Rate: High 40.2, Medium 28.2, Low 31.6; 3.17 ± 1.8
Scenario 2	Alternate_Income_Opportunity: Yes 70.0, No 30.0	Anxiety: Yes 70.0, No 30.0	Payment_Delay: Yes 70.0, No 30.0	Absenteeism_Rate: High 38.4, Medium 28.4, Low 33.2; 3.1 ± 1.8
Scenario 3	Alternate_Income_Opportunity: Yes 50.0, No 50.0	Anxiety: Yes 50.0, No 50.0	Payment_Delay: Yes 50.0, No 50.0	Absenteeism_Rate: High 36.6, Medium 28.5, Low 34.9; 3.03 ± 1.8
Scenario 4	Alternate_Income_Opportunity: Yes 30.0, No 70.0	Anxiety: Yes 30.0, No 70.0	Payment_Delay: Yes 30.0, No 70.0	Absenteeism_Rate: High 34.9, Medium 28.4, Low 36.6; 2.97 ± 1.8
Scenario 5	Alternate_Income_Opportunity: Yes 10.0, No 90.0	Anxiety: Yes 10.0, No 90.0	Payment_Delay: Yes 10.0, No 90.0	Absenteeism_Rate: High 33.3, Medium 28.3, Low 38.4; 2.9 ± 1.8

Table 14
Result of scenario analysis on Model I & II.

Scenario	Alternate Income Opportunity	Anxiety	Payment Delay	Absenteeism Risk (%)	Absenteeism Rate
1	0.9	0.9	0.9	55.2	3.17
2	0.7	0.7	0.7	52.9	3.10
3	0.5	0.5	0.5	50.5	3.03
4	0.3	0.3	0.3	48.2	2.97
5	0.1	0.1	0.1	45.9	2.90

been observed that they often function based on the assumptions of linearity and the presence of complete datasets [35]. The challenges posed by limited data availability are addressed through the utilisation of expert opinion within the Bayesian Belief Network framework. Functional interpolation is utilized to alleviate the demands of probability elicitation, while Monte Carlo Simulation is incorporated to effectively measure uncertainty. This comprehensive approach ensures that the predictions are rendered more resilient and are accurately reflective of the complexities inherent in real-world organizational environments, in which data is frequently encountered in an incomplete or noisy state.

The findings from the case study of the garment manufacturing sector in Bangladesh offer valuable insights that contribute to the existing body of literature. Initially, the anticipated average absenteeism rate of 2.49 days per month aligns with international studies indicating medium-range absenteeism in industrial sectors. Additionally, it draws

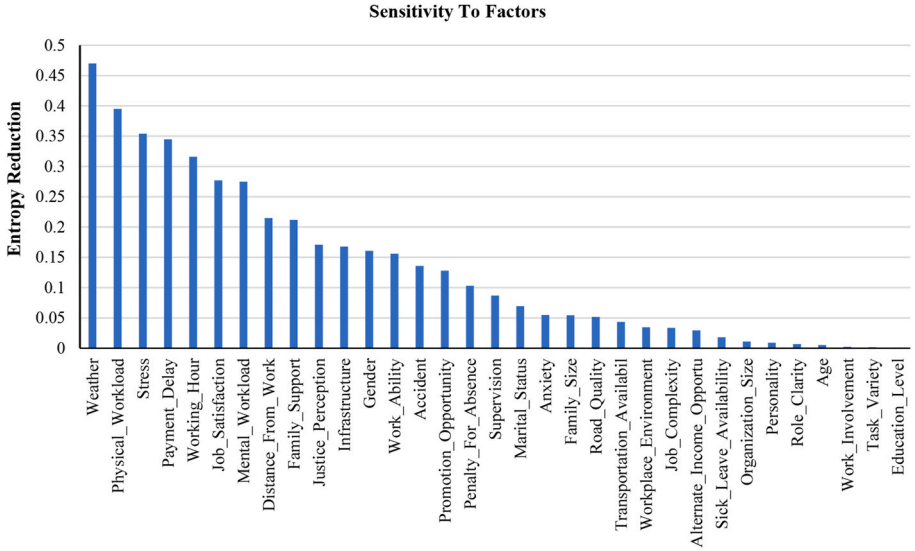


Fig. 7. Result of sensitivity analysis (type I) for Model I.

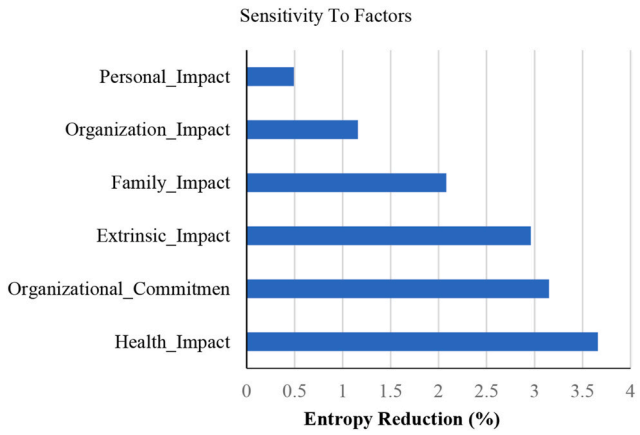


Fig. 8. Result of sensitivity analysis (type II) for Model I.

attention to the significant percentage of employees experiencing high absenteeism (23.9%), a distributional perspective that has been infrequently addressed in previous research. The ability to deliver predictions at both the individual and section levels represent a significant methodological improvement compared to previous models, which were constrained to either micro-level (focused on individuals) or macro-level (organization-wide) assessments. Furthermore, the sensitivity analysis indicated that weather accounted for 36.2% of the reduction in entropy, establishing it as the most significant factor. Although previous studies have recognized the role of environmental factors [32], these elements have frequently been eclipsed by a focus on organizational or health-related influences. Our findings indicate that extrinsic and environmental factors may hold greater significance than previously recognized, especially in labor-intensive industries where infrastructure and climate have a direct impact on daily attendance. Third, the validation process revealed that the models exhibit consistent performance across extreme scenarios, stochastic variations, and diagnostic tests. This highlights the models' ability to be applied broadly and their usefulness as decision-support tools for managers, particularly in environments with limited resources.

In addition to recognizing the factors that contribute to absenteeism, research in this area is placing greater importance on preventive measures [68]. highlighted how vaccination significantly contributed to lowering both absenteeism rates and the duration of absences among healthcare workers during and following the COVID-19 pandemic. While vaccination was not explicitly part of the current dataset, the BBN framework demonstrates a remarkable capacity for adaptation. Preventive measures, including vaccination, workplace health campaigns [69], stress management programs, and ergonomic interventions, can be incorporated as new primary nodes within the health-related category. Implementing such measures would enable organizations to explore various "what-if" scenarios, thereby estimating the potential decrease in both the risk and rate of absenteeism that could result from these interventions. For instance, integrating vaccination into the health-related nodes may influence the probabilities associated with "work ability" and "disease occurrence," consequently reducing the risk of absenteeism throughout the network. In a similar vein, wellness programs have the potential to impact areas related to stress and anxiety, while policies that allow for flexible scheduling may help reduce absenteeism linked to family responsibilities. Through the explicit integration of preventive measures, the BBN approach transcends mere prediction, enabling proactive management of absenteeism. This provides organizations with a valuable tool for not only comprehending current risks but also for anticipating the advantages of policy interventions.

When considered collectively, these findings both affirm and broaden the existing body of literature in multiple aspects. The authors emphasize that absenteeism is influenced by a diverse array of factors identified in earlier research. Furthermore, they illustrate that by considering uncertainty, nonlinear interactions, and preventive strategies, one can achieve a more nuanced and applicable comprehension of the issue. The BBN framework marks an important advancement in methodology, allowing organizations to assess absenteeism in a comprehensive manner, model potential interventions, and customize strategies to fit the unique contexts of their workforce.

7. Conclusions, limitations, and future research directions

Employee absenteeism has been acknowledged as a significant managerial challenge, with its influence extending across various dimensions of organizational performance. In this study, two forecasting models based on Bayesian Belief Networks (BBN) were developed to predict the risk and rate of absenteeism among employees.

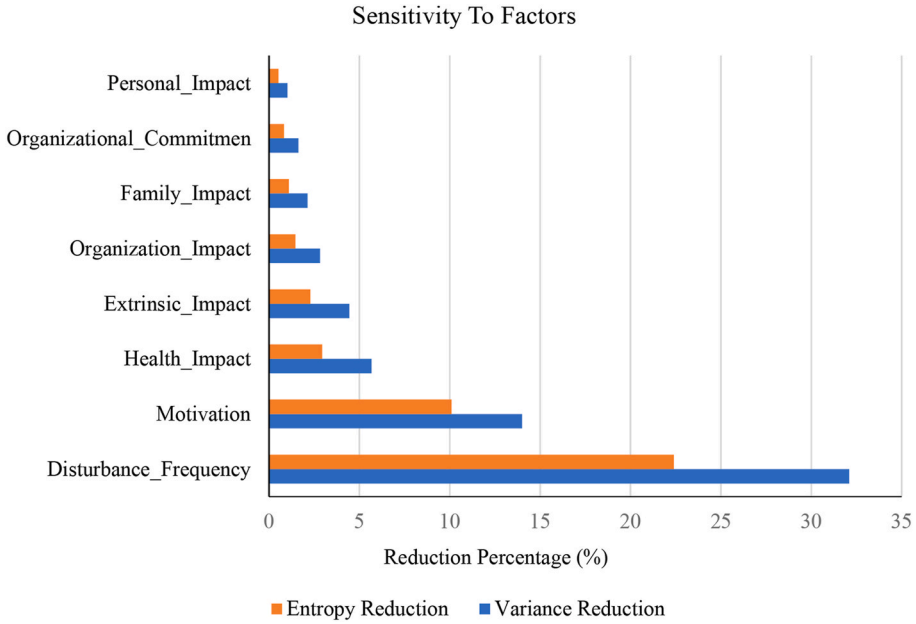


Fig. 9. Result of sensitivity analysis (type II) for Model II.

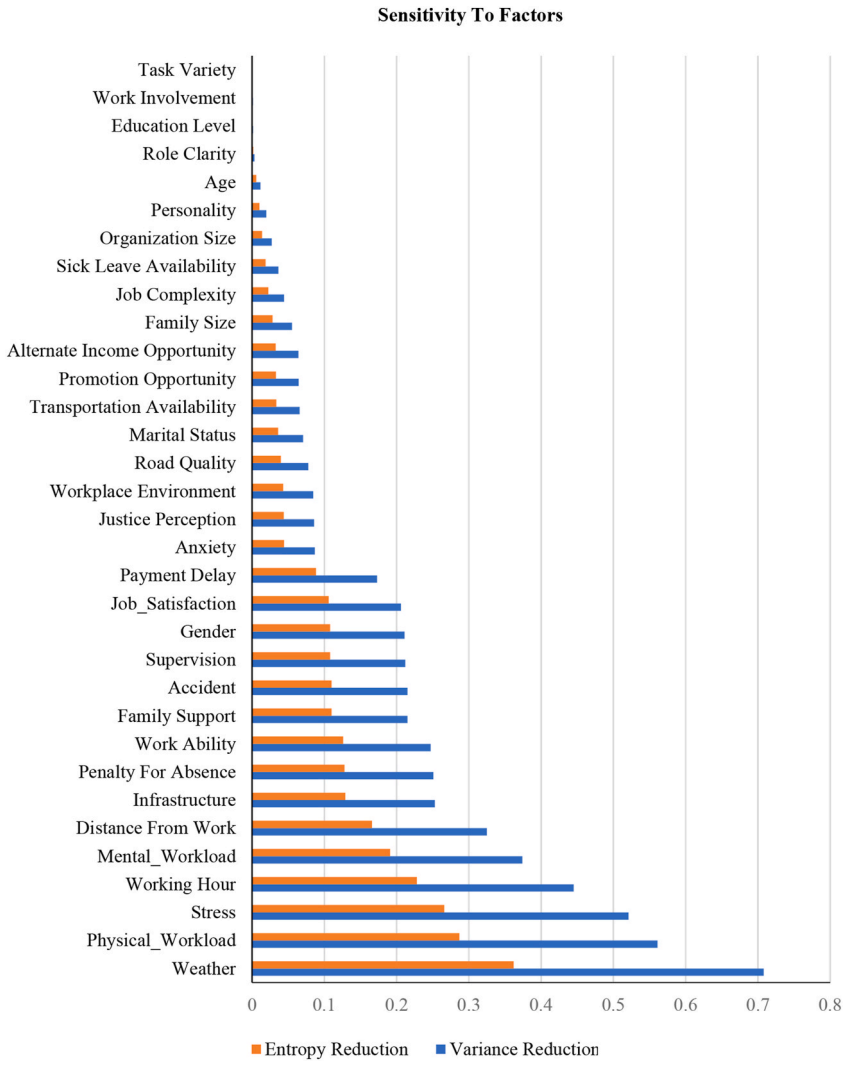


Fig. 10. Result of sensitivity analysis (type I) for Model II.

This research is characterized by the formulation of probabilistic models that facilitate the prediction of absenteeism through the monitoring of a selected set of indicators. The proposed framework allows for the estimation of the attendance probability of individual employees, along with the mean absence rate of the entire workforce. It is suggested that the insights derived from such estimations could aid organizations in the adoption of preventive measures and the development of screening procedures during recruitment. The identification of the most influential variables related to absenteeism is anticipated to improve the efficiency of data analytics in studies concerning absence.

A simplified procedure for expert elicitation was developed to aid in the construction of Conditional Probability Tables (CPTs). By employing this approach, it was ensured that expert opinions were incorporated without the necessity of precise numerical probabilities, which contributed to a reduction in cognitive burden and an enhancement of practicality. The workload of domain specialists was further minimized through the use of functional interpolation, allowing for the modeling process to be feasible even in situations where data are limited. The model was tailored to meet organizational needs through the employment of factor analysis, which contributed to its enhanced adaptability and robustness. The implementation of Monte Carlo Simulation (MCS) was subsequently undertaken to incorporate uncertainty, thereby rendering the predictions more realistic. The integration of multiple expert opinions was facilitated by the BBN framework, and the resulting probability effects on absenteeism were quantitatively assessed. The ability of these models to incorporate new evidence is such that prediction accuracy may be enhanced as further data are made available.

Although these contributions were acknowledged, it was noted that several limitations exist. The development and validation of the models were conducted utilizing data from Bangladeshi industries; consequently, it is acknowledged that their parameters may exhibit variation across diverse socio-economic or cultural contexts. It would thus be necessary for model recalibration to occur in order to facilitate replication in other regions. The CPTs were generated using a rank-sum elicitation technique, whereby the influence of each factor was evaluated independently. Nevertheless, the consideration of equal or interdependent effects among multiple factors was not undertaken. Furthermore, it has been noted that the functional interpolation utilized a normal distribution, which might not be considered universally optimal; it is suggested that alternative ranking systems and probability distributions be explored in future research. Given that the CPTs were derived through subjective means, it is acknowledged that discrepancies may arise between expert judgment and real-world behavior. Nevertheless, a mechanism for continual refinement through evidence-based updating is provided by the inherent learning capability of BBNs. Ultimately, the simplification of interrelationships among nodes was undertaken to reduce the complexity of the model, which may have resulted in constraints on analytical precision.

It is ultimately recognized that direct collaboration with behavioral researchers was not feasible during the initial phase of this study, owing

to constraints related to scope and resources. It is acknowledged that interdisciplinary collaboration is viewed as a promising avenue for future research. Through the integration of expertise from behavioral science and organizational psychology, it is anticipated that subsequent studies will refine the psychological constructs embedded in the BBN, validate them empirically through behavioral experiments, and thereby strengthen the model's theoretical grounding and interpretive validity.

It is suggested that future research could address these limitations through the expansion of the dataset to encompass diverse industries and geographical regions, experimentation with alternative elicitation techniques and distributional assumptions, and the incorporation of inter-node dependencies to achieve higher accuracy. Additionally, the investigation of dynamic or hybrid BBN models may be conducted to capture temporal variations in absenteeism behavior, which could lead to enhancements in predictive reliability and managerial relevance.

CRediT authorship contribution statement

Rashid Anzoom: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Syed Mithun Ali:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Prattusha Bhattacharjee:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Golam Kabir:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Sanjoy Kumar Paul:** Writing – review & editing, Visualization, Methodology, Conceptualization.

Ethical approval

Not applicable.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used QuilBot Premium/Grammarly Premium/ChatGPT to improve the quality of the writing and check for any grammatical errors. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.array.2026.100766.

Data availability

Data used are found in the paper and in supplementary materials.

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